

Definite integrals – 3

Antiderivatives and the Newton–Leibniz formula

Homework

8 сентября 2026 года

Theoretical minimum. The integral with a variable upper limit and its Lipschitz property; differentiability of F at continuity points; the Newton–Leibniz formula and its two independent hypotheses; the Darboux property of derivatives; Volterra’s function; integration by parts and change of variables with honest hypotheses.

1. (RMO) Let $\mathcal{F}$ be the set of functions $f: \mathbb{R} \rightarrow \mathbb{R}$ with $f(2x) = f(x)$ for all x .
 - a) Determine all $f \in \mathcal{F}$ which admit antiderivatives on $\mathbb{R}$.
 - b) Give a non-constant $f \in \mathcal{F}$ which is integrable on every $[a, b]$ and satisfies $\int_a^b f = 0$ for all $a < b$.
2. Prove the formulas of integration by parts and of change of variables for the Riemann integral, stating the exact hypotheses, and give an example where a non-monotone substitution produces a wrong answer if the hypotheses are ignored.
3. Prove that if f is integrable and $\int_a^x f = 0$ for all $x \in [a, b]$, then f vanishes at every point of continuity; and that if f is continuous then $f \equiv 0$.
4. Prove that if f' exists at every point of $[a, b]$ and f' is Riemann integrable, then $\int_a^b f' = f(b) - f(a)$; then show that the conclusion may fail if f' is only assumed to exist outside a countable set, even for a continuous f .
5. (RMO) Say that $f: [a, b] \rightarrow \mathbb{R}$ has property (P) if there is an integrable function such that $f(x) - f\left(\frac{x+a}{2}\right) = f\left(\frac{x+b}{2}\right) - f(x)$ for all x . Prove that for every real t there is exactly one f with property (P) and $\int_a^b f = t$.

6. Prove that $F(x) = \int_a^x f$ has bounded variation and that its total variation over $[a, b]$ equals $\int_a^b |f|$. Deduce that F is monotone on an interval if and only if f does not change sign there in the essential sense.
7. Let f be continuous and T -periodic. Prove that $\int_a^{a+T} f$ does not depend on a , and that $F(x) = \int_0^x f$ is T -periodic if and only if $\int_0^T f = 0$. Describe F in general as a linear function plus a periodic one.
8. Prove that a continuous function which is differentiable outside a finite set and whose derivative extends continuously satisfies the Newton–Leibniz formula. Then explain why the Cantor function, built in Continuous functions – 8, shows that the formula fails without further assumptions, and say exactly which hypothesis of Theorem 2 it violates.
9. Let f be integrable on $[0, 1]$. Prove that $\lim_{h \rightarrow 0^+} \frac{1}{h} \int_x^{x+h} f = f(x)$ at every point of continuity, and construct an integrable f and a point x where the limit does not exist.
10. Prove that a bounded f with an antiderivative on $[a, b]$ is integrable if and only if for every $\sigma > 0$ the set $\{x : \omega(f, x) \geq \sigma\}$ can be covered by finitely many intervals of arbitrarily small total length. Illustrate both possibilities: $\sin \frac{1}{x}$ on one side, and on the other the derivative built in Differentiable functions – 8.